

Silver Plating + Faraday's Law

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Estimate how many atoms were deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 3 — Silver Plating and Faraday's Law

Hook: Calculate how many silver atoms you deposit and how thick the film is.

Learning outcomes

- Plate silver onto a metal object with controlled current and time
- Use $Q = It$, $n = Q/F$, and mass/thickness relations
- Discuss assumptions (100% efficiency, uniform coverage)
- Compare ideal Faraday prediction to visual result

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: dull copper object — can we count the atoms we add?	lecture.md
10–25 min	Ag^+ reduction and Faraday's law	lecture.md
25–35 min	Estimate plated surface area	experiment.md — Area
35–45 min	Set up cell with ammeter in series	experiment.md — Setup
45–65 min	Plate with timed current readings	experiment.md — Run
65–80 min	Calculate charge, moles, mass, thickness	experiment.md — Calculations
80–90 min	Discuss efficiency and side reactions	lecture.md

Key equations

$$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$$

$$Q = I \times t$$

$$n_{\text{e}} = Q / F \quad (F = 96,485 \text{ C/mol})$$

$$n_{\text{Ag}} = n_{\text{e}}$$

$$m_{\text{Ag}} = n_{\text{Ag}} \times M_{\text{Ag}} \quad (M_{\text{Ag}} = 107.87 \text{ g/mol})$$

$$d = m_{\text{Ag}} / (\rho_{\text{Ag}} \times A) \quad (\rho_{\text{Ag}} = 10.49 \text{ g/cm}^3)$$

Status

- ☐ AgNO_3 solution prepared (0.01 M)
- ☐ Calculation worksheet ready
- ☐ Pilot plating run completed
- ☐ Waste container labeled for silver

Session 3 — Lecture: Faraday's Law and Silver Plating

Target duration: ~20 minutes

Instructors: worked Faraday example, z-factor for Ag vs Cu, efficiency — instructor.md.

Opening hook (0–10 min)

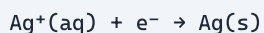
Show a dull copper coin or washer.

"If we plate this with silver for exactly 5 minutes at 0.02 A, can we calculate how many silver atoms landed on the surface — without weighing it?"

Answer preview: **Yes — with Faraday's law.**

Core concepts (10–25 min)

1. Silver reduction



- Each silver ion needs **one electron** to become silver metal
- So: **moles of electrons = moles of silver deposited** (1:1)

2. Charge, current, and time

Symbol	Meaning	Units
I	Current	Amperes (A) = C/s
t	Time	Seconds (s)
Q	Charge	Coulombs (C)

$$Q = I \times t$$

Example: 0.020 A for 300 s $\rightarrow$ Q = 6.0 C

3. Faraday constant

F = 96,485 C/mol — charge of one mole of electrons

$$n_{\text{e}} = Q / F$$

Example: 6.0 / 96,485 = 6.22×10^{-5} mol e^-

4. From moles to mass to thickness

1. $n_{\text{Ag}} = n_{\text{e}}$ (1 electron per Ag^+)

2. $m_{\text{Ag}} = n_{\text{Ag}} \times 107.87 \text{ g/mol}$
3. Volume $V = m_{\text{Ag}} / \rho_{\text{Ag}}$ ($\rho_{\text{Ag}} = 10.49 \text{ g/cm}^3$)
4. Thickness $d = V / A$ (A = plated area in cm^2)

5. Worked example (on board)

Step	Calculation	Result
Q	$0.020 \text{ A} \times 300 \text{ s}$	6.0 C
n_e	$6.0 / 96,485$	$6.22 \times 10^{-5} \text{ mol}$
m_{Ag}	$n \times 107.87$	0.00671 g
V	$m / 10.49$	$6.40 \times 10^{-4} \text{ cm}^3$
d ($A = 10 \text{ cm}^2$)	V / A	$6.4 \times 10^{-5} \text{ cm} \approx \mathbf{0.64 \text{ }\mu\text{m}}$

6. Ideal vs. real plating

Faraday gives the **maximum** if every electron plates silver.

Reality may differ because:

- Side reactions consume current (H_2 evolution, etc.)
- Deposit may be porous or non-uniform
- Current may drift over time
- Not all surface area plates equally

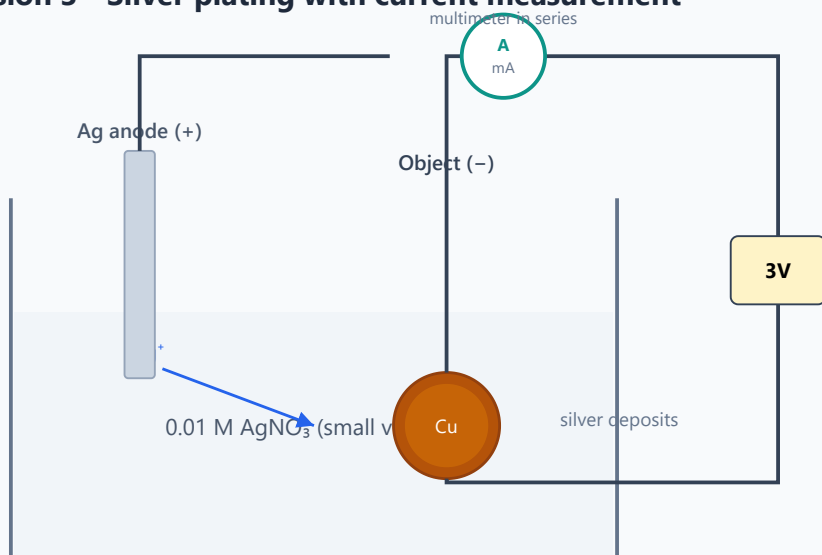
Wrap-up (80–90 min)

- How close was your calculated thickness to what you see?
- What would happen with a **silver anode** vs. inert anode?
- Where else is Faraday's law used? (electrolysis industry, battery design, analytical chemistry)

Visual aids

- [] Flowchart: $I, t \rightarrow Q \rightarrow n \rightarrow m \rightarrow d$
- [x] Plating setup (see experiment page and figure below)

Session 3 - Silver plating with current measurement



$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag(s)}$ Record I every 30-60 s and total time t

$Q = I \times t \rightarrow n = Q/F \rightarrow m_{\text{Ag}} \rightarrow \text{thickness } d = m/(\rho A)$

Estimate plated area A before starting (coin face, washer ring, etc.)

- [] Optional: stacked-atom illustration for "thickness"

Vocabulary

- [] Faraday constant
- [] Coulomb
- [] Stoichiometry of electron transfer
- [] Current efficiency (introductory)

Bridge to Session 4

"We've moved metal atoms with electricity. Next we split water molecules into hydrogen and oxygen — and measure the gases to prove the 2:1 ratio."

Session 3 — Instructor Notes: Silver Plating + Faraday Deep Dive

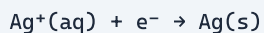
Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Students saw copper move in Session 2. Now they **quantify** electrodeposition: charge → moles of electrons → moles of metal → mass → thickness. Silver is ideal because $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ is 1:1 (simpler stoichiometry than $\text{Cu}^{2+} + 2\text{e}^-$). This is the week's heaviest math session — go slowly on units.

Atomic / electronic foundations for silver

Silver ion and electron count



- Silver atom (Ag, $Z = 47$) loses one valence electron → Ag^+ .
- Plating reverses that: one electron returns → one Ag atom deposited.
- Therefore: **$n_{\text{Ag}} = n_{\text{e}}$** (moles silver = moles electrons) when plating is 100% efficient.

Compare with copper from Session 2:



Mention this contrast so students see why the electron count in the half-reaction matters.

Why dilute AgNO_3

Silver nitrate provides Ag^+ . Classroom concentration **0.01 M** keeps:

- Cost down
- Staining / hazard manageable
- Thin films that still look metallic

Safety (non-negotiable): AgNO_3 stains skin/clothes permanently dark; gloves + goggles; collect all silver waste.

Faraday's law — instructor mastery

Charge from current and time

$$Q = I \times t$$

Symbol	Unit	Meaning
I	ampere (A) = C/s	Current
t	second (s)	Time
Q	coulomb (C)	Total charge passed

Trap: students record mA but forget to convert. 20 mA = 0.020 A.

Faraday constant

$$F = 96,485 \text{ C/mol } e^- \approx 9.65 \times 10^4 \text{ C/mol}$$

Meaning: one mole of electrons carries ~96,485 C.

$$n_e = Q / F$$

Stoichiometry factor (general form)

For a reduction requiring **z** electrons per metal ion:

$$n_{\text{metal}} = n_e / z = Q / (z F)$$

Metal ion	z	Example
Ag ⁺	1	Session 3
Cu ²⁺	2	Session 2
Al ³⁺	3	not used here

Mass and thickness

$$m = n \times M$$

$$V = m / \rho$$

$$d = V / A$$

Quantity	Silver value
Molar mass M	107.87 g/mol
Density ρ	10.49 g/cm ³
A	geometric plated area (cm ²) — biggest student error source

Worked example (memorize this board version)

Given: I = 0.020 A, t = 300 s, A = 10 cm², Ag plating, z = 1.

Step	Math	Result
Q	0.020×300	6.0 C
n_e	$6.0 / 96485$	$6.22 \times 10^{-5} \text{ mol}$
n_Ag	$= n_e$	$6.22 \times 10^{-5} \text{ mol}$
m	$6.22 \times 10^{-5} \times 107.87$	$6.71 \times 10^{-3} \text{ g}$
V	$m / 10.49$	$6.40 \times 10^{-4} \text{ cm}^3$
d	$V / 10$	$6.4 \times 10^{-5} \text{ cm} = \mathbf{0.64 \mu m}$

Sense-check: visible silver films are often tenths of a micrometer to a few micrometers — students' numbers should be in that ballpark if I, t, A are sane.

Average current when I drifts

If students log I every 30–60 s:

$$I_{\text{avg}} \approx (I_0 + I_1 + \dots + I_n) / (n+1)$$

or trapezoidal estimate. Then $Q \approx I_{\text{avg}} \times t_{\text{total}}$.

Do **not** use only the first reading if current falls (common as concentration polarization / surface changes).

Ideal vs real plating (current efficiency)

Faraday assumes **every** electron reduces Ag^+ . Reality:

$$\text{current efficiency } \eta = (\text{charge used for Ag plating}) / (\text{total charge}) \leq 1$$

Loss channels:

- H_2 evolution
- Impurity reductions
- Non-uniform current (edges plate more — “dog-boning”)
- Flaky deposit that rinses off

If calculated thickness looks “too thick” vs visual: area underestimate or optimistic $\eta = 1$. If “too thin”: efficiency < 1 , area overestimate, or current logged wrong.

Electrochemical series note for Ag

Couple	E° (V)
Ag^+/Ag	+0.80
Cu^{2+}/Cu	+0.34

Couple	E° (V)
H ⁺ /H ₂	0.00

Ag⁺ is a strong oxidant relative to H⁺ and Cu²⁺ — silver plates readily. That is why even dilute AgNO₃ works visually. It is also why silver ions are “eager” to reduce and why waste must be handled carefully (and why Ag stains via reduction on skin/organics).

Anode choice:

- **Silver anode:** replenishes Ag⁺ (like Cu anode in Session 2).
- **Inert anode:** Ag⁺ depletes; oxygen evolution possible; calculations still OK for short runs if solution starts with enough Ag⁺.

Experiment coaching

Area estimation — coach aggressively

Object	Reasonable approach
Coin face	$A = \pi(r)^2$ per face; $\times 2$ if both faces plate
Washer	$\pi(R^2 - r^2)$
Irregular	Length $\times$ width of immersed region; say so

Students who invent a huge area get absurdly small thickness and think Faraday “failed.”

Ammeter in series

Multimeter must be **in series** on DC current range — not across the cell like a voltmeter (that shorts / misreads). Walk each station once.

Timing discipline

Start stopwatch when current is flowing and stable. Stop when disconnected. Gaps in logging → poor I_{avg} .

What success looks like

- Whitish / silvery sheen on copper object
- Not necessarily mirror-bright (classroom baths lack brighteners)
- Calculation completed with units at every step

Misconceptions

Misconception	Correction
“ $Q = I + t$ ”	$Q = I \times t$
“Use mA directly in $Q = It$ ”	Convert to amperes

Misconception	Correction
"Cu and Ag use the same z"	Ag^+ needs 1e^- ; Cu^{2+} needs 2e^-
"Thickness must match a ruler"	Micrometers are invisible to a ruler; appearance is qualitative
"Faraday is wrong if deposit looks thin"	Efficiency and area errors dominate

Board plan

1. Write $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$; circle "1 electron per atom."
2. Flowchart: $I, t \rightarrow Q \rightarrow n_{\text{e}} \rightarrow n_{\text{Ag}} \rightarrow m \rightarrow V \rightarrow d$.
3. Full numerical example with units.
4. Discuss why I_{avg} beats a single reading.
5. Bridge to Session 4: next we count **gas molecules** by volume ratio, not metal atoms by charge.

Optional enrichment

- Derive z from half-reaction for Cu plating retroactively (Session 2 data).
- Mention electrolysis mass production (Al from Hall–Héroult — huge Faraday consumer).
- Introduce faradaic efficiency formally if a student is ready.

Session 3 — Preparation

Before class (instructor)

- ☐ Prepare 0.01 M AgNO_3 with **distilled/deionized water** (not tap); volume for all groups + 20% extra
- ☐ Pilot plate one object: record I_{avg} , t , appearance
- ☐ Verify 3 V packs deliver stable $\sim 10\text{--}30$ mA in this cell
- ☐ Print calculation worksheets
- ☐ Confirm silver waste disposal with school/facility protocol

AgNO_3 handling

- Wear gloves when preparing
- Work in ventilated area
- Label all bottles clearly
- Store away from organic materials and light if possible

Day-of setup

- ☐ Small volumes pre-measured per station (reduces spills)
- ☐ Goggles enforced at door
- ☐ Waste container visible and labeled
- ☐ Demo calculation on board with example numbers (before students start)

Timing note

Calculations take longer than expected for some groups. Have I_{avg} pre-computed on board as check, or provide a spreadsheet template.

Open questions / notes

- Stock AgNO_3 concentration on hand:
- Volume prepared (mL):
- Pilot I_{avg} (mA):
- Pilot plating time (s):